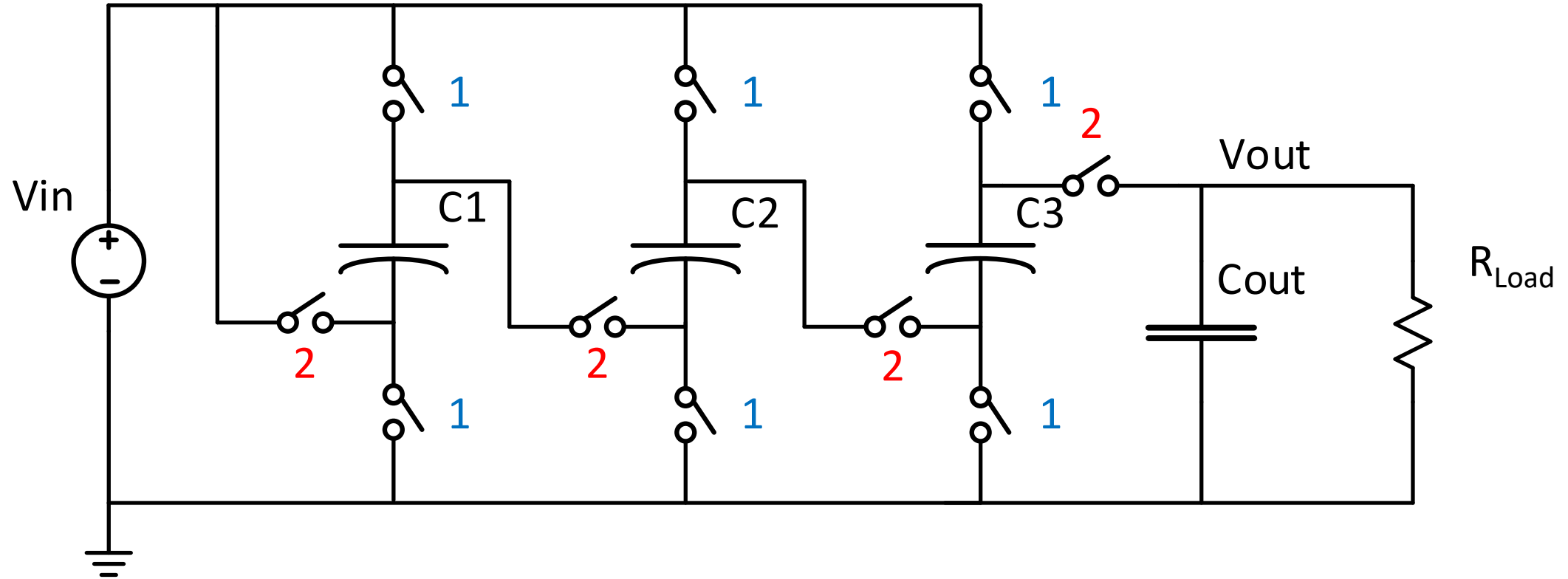


ET4382: Power Conversion Techniques

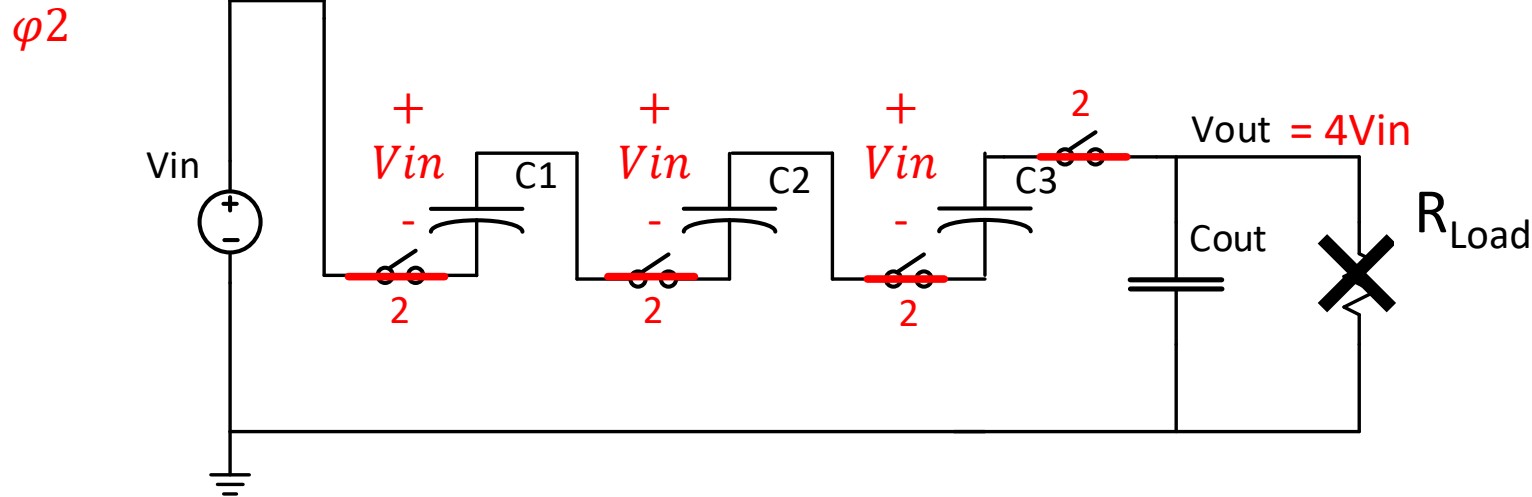
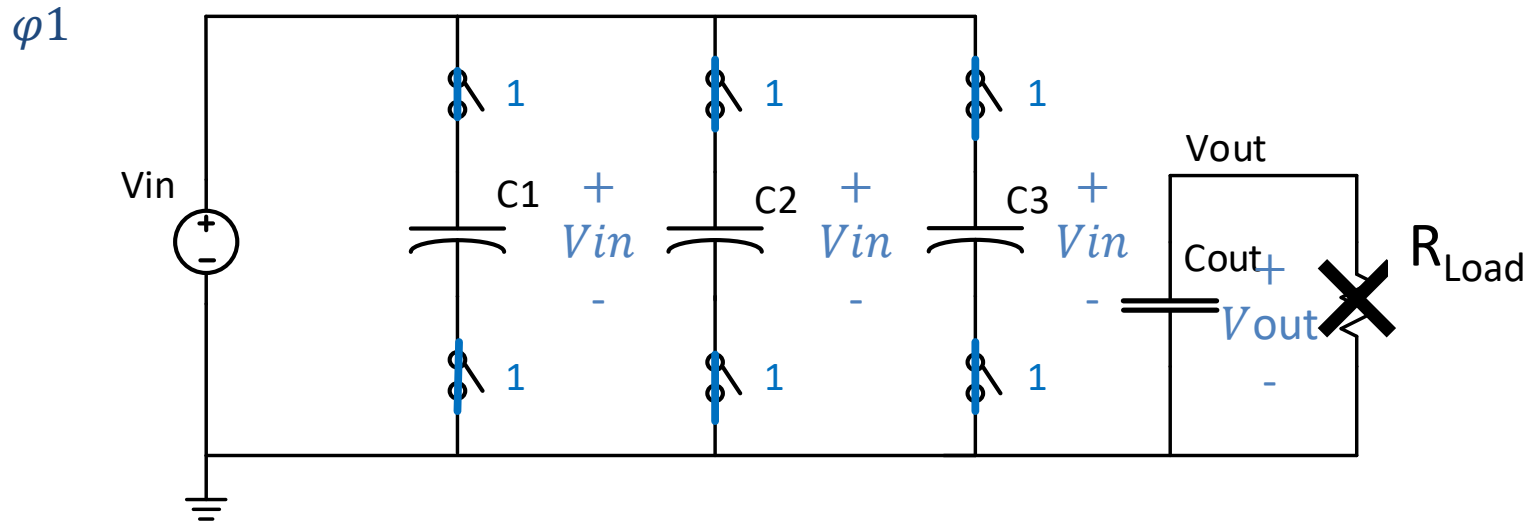
Tutorial 6

18th March, 2021

Q1) What is the conversion ratio of this power converter?

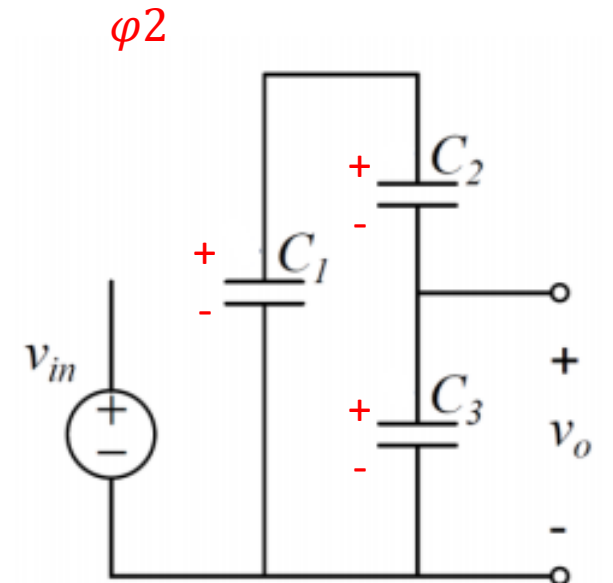
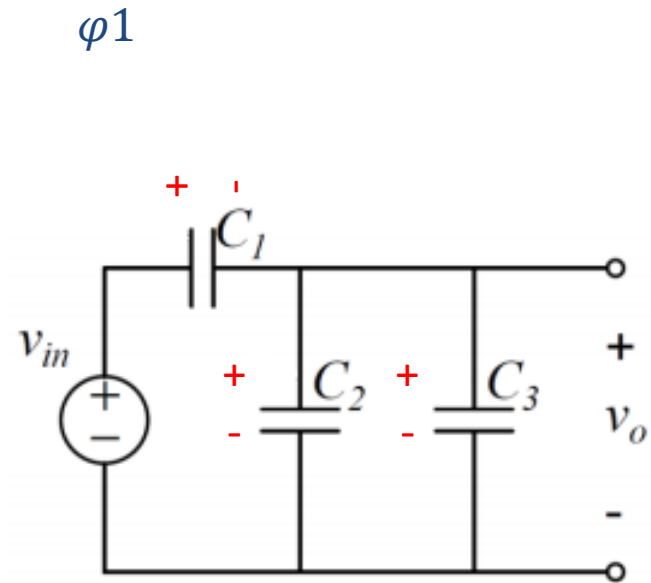
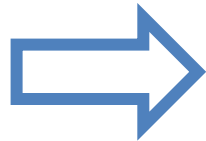
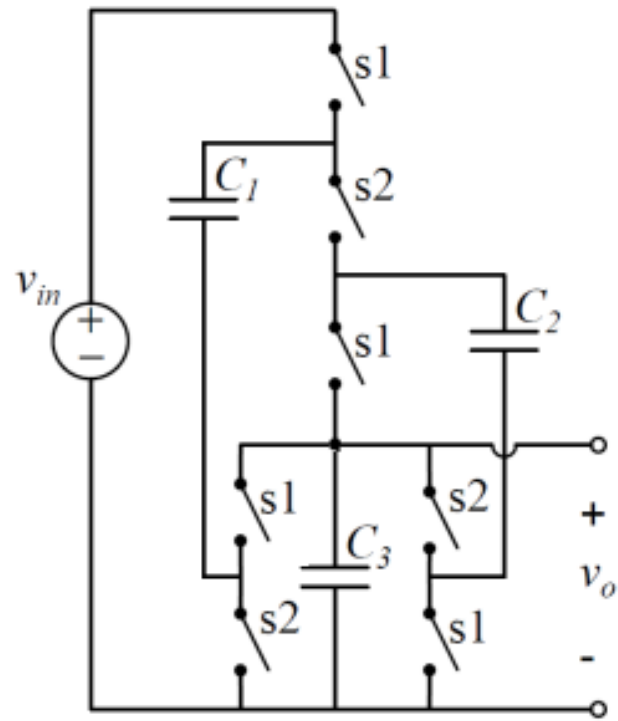


A1) To find voltage conversion ratio M , assume $I_{out}=0$.



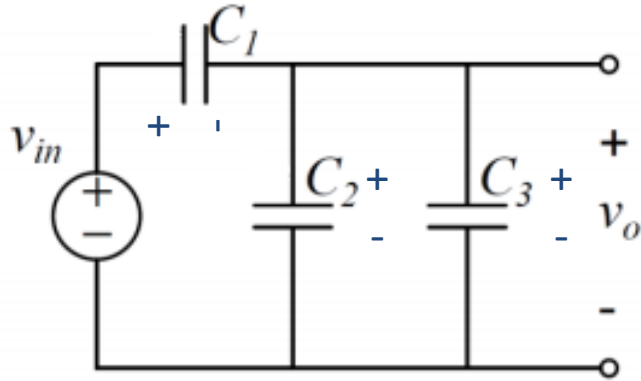
M=4 Series-Parallel SCPC

Q2) What is the conversion ratio of this power converter?



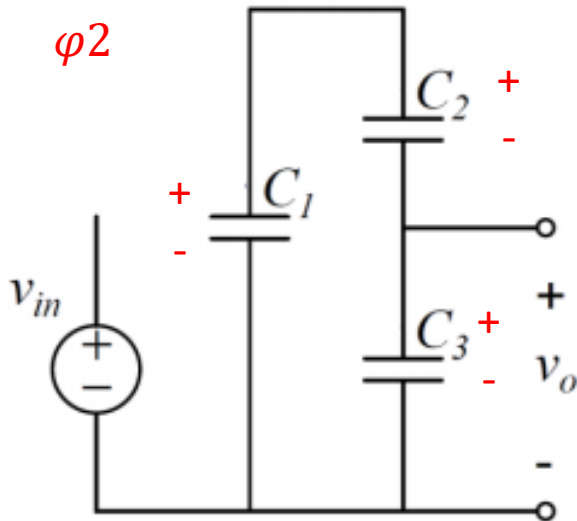
A2) To find voltage conversion ratio M , assume $I_{out}=0$.

$\varphi 1$



$$\begin{aligned}\varphi 1: \quad V_o &= V_{C3} = V_{C2} \\ V_{C1} &= V_{in} - V_o\end{aligned}$$

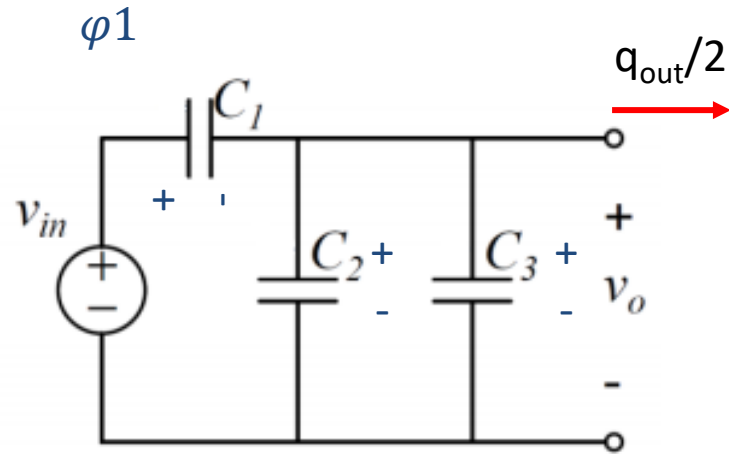
$\varphi 2$



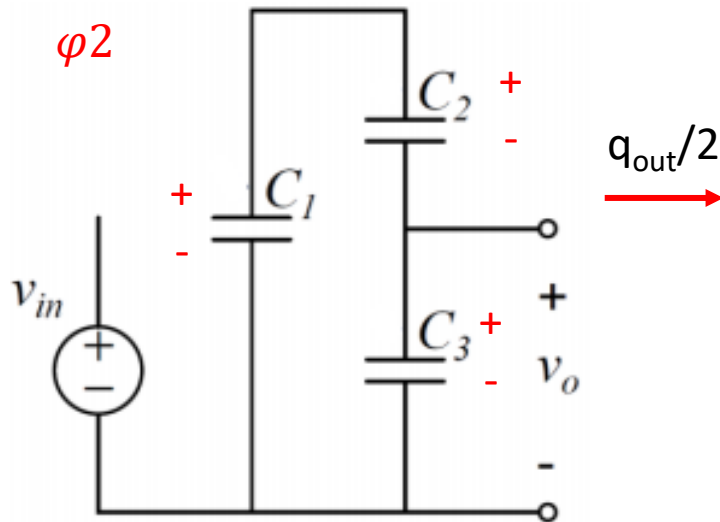
$$\begin{aligned}\varphi 2: \quad V_o &= V_{C1} - V_{C2} = (V_{in} - V_o) - \frac{V_{in}}{3} \\ V_o &= \frac{V_{in}}{3}\end{aligned}$$

$M=1/3$ Dickson SC Converter

Q3) How much charge flows through each capacitor in terms of q_{out} ?

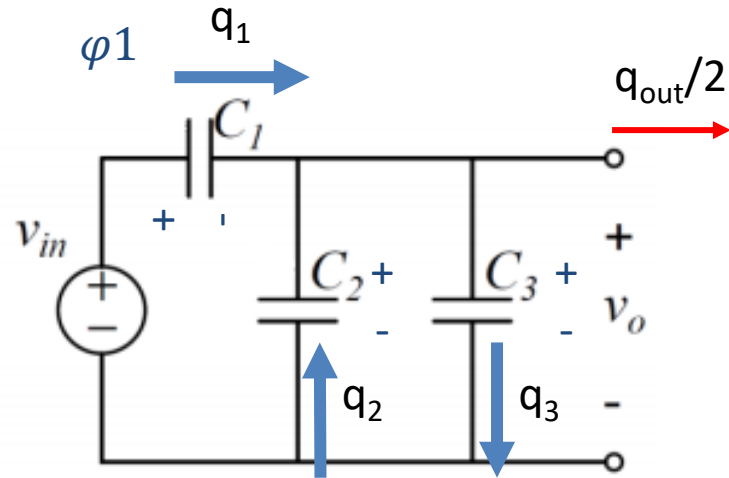


$$D = 50\% \Rightarrow q_{out, \varphi 1} = q_{out, \varphi 2} = q_{out}/2$$

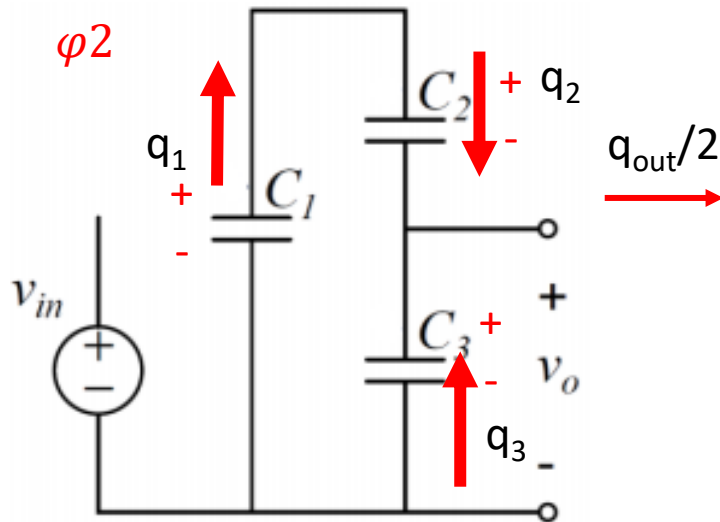


M=1/3 Dickson SC Converter

A3) How much charge flows through each capacitor in terms of q_{out} ?



$$\phi 1: \quad q_1 = \frac{q_{out}}{2} - q_2 + q_3$$



$$\phi 2: \quad q_1 = q_2$$

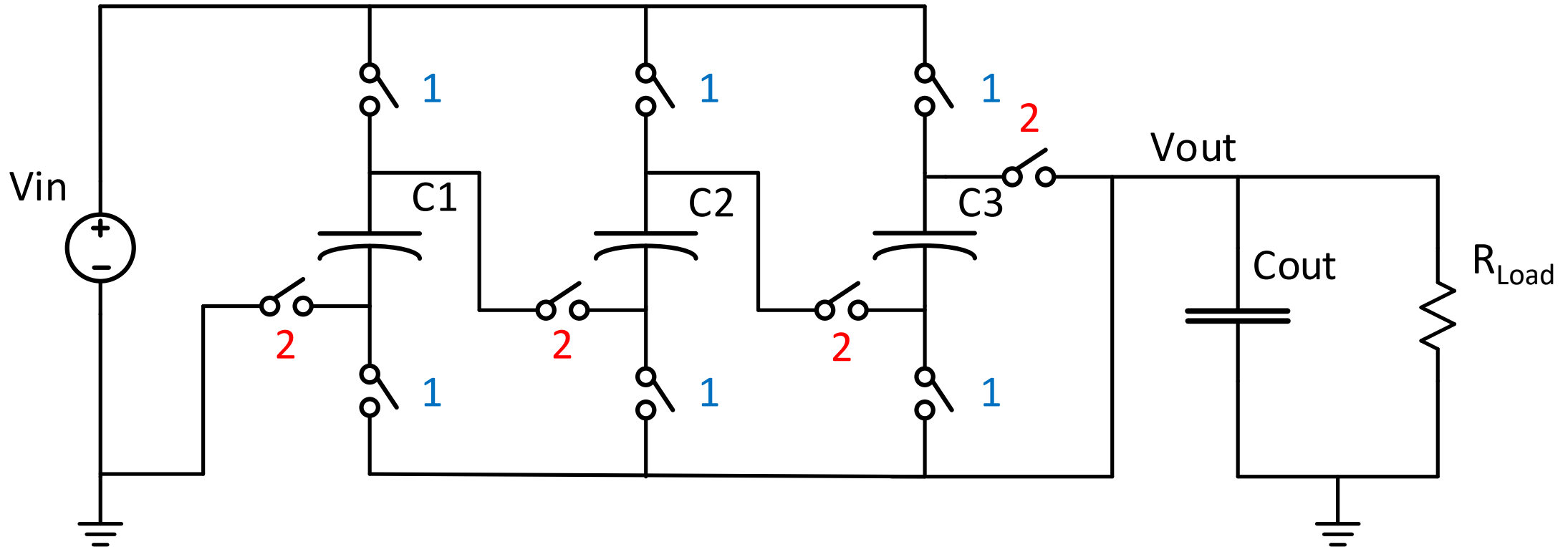
$$q_1 = \frac{q_{out}}{2} - q_3$$

$$2q_1 = \frac{q_{out}}{2} + q_3$$

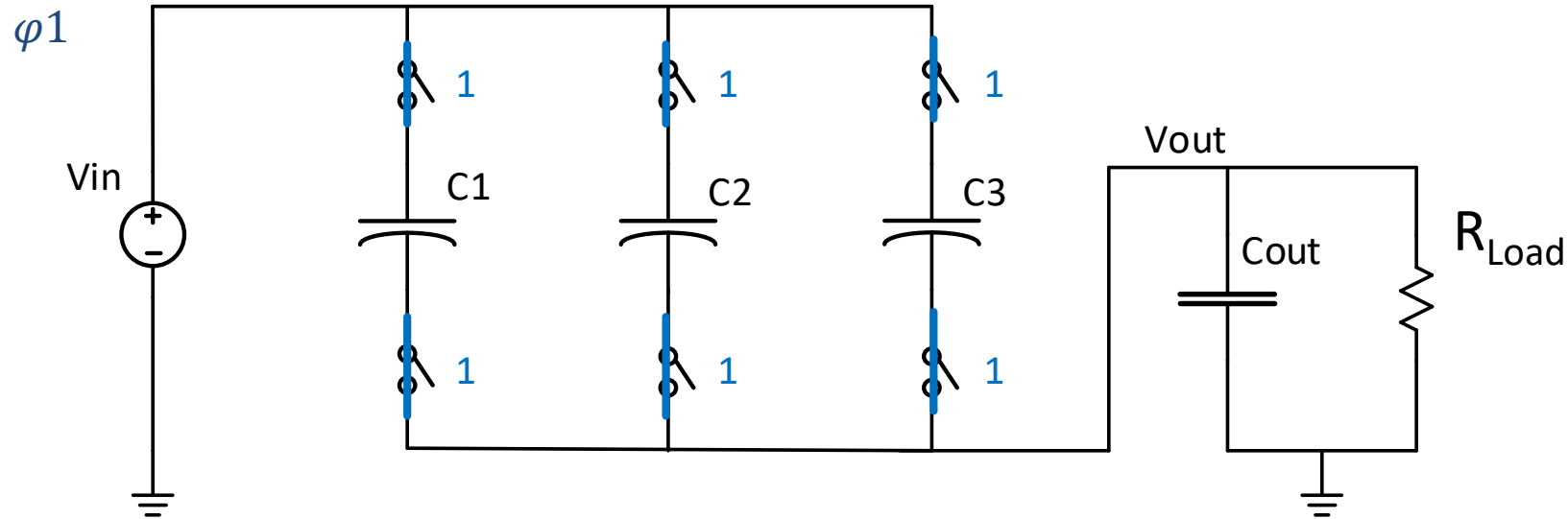
$$q_1 = q_2 = q_{in} = q_{out}/3$$

$$q_3 = q_{out}/6$$

Q4) What is the conversion ratio of this power converter?

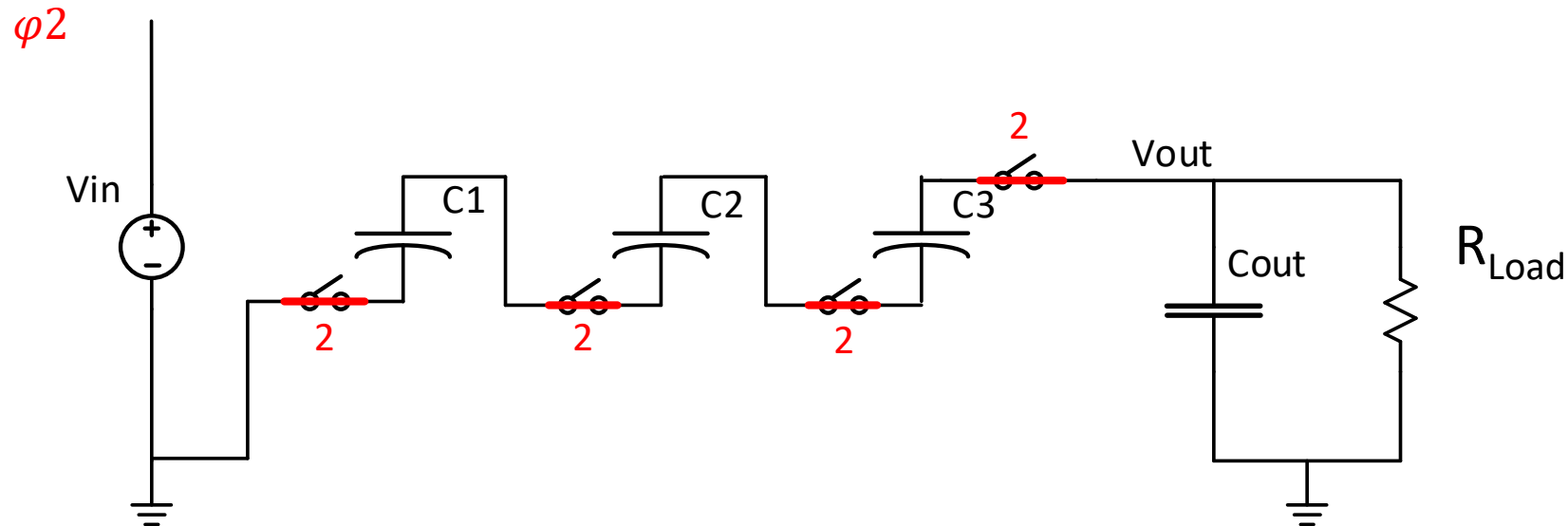


A4) Assume $I_{out}=0$, to find voltage conversion ratio M .



$$\phi 1: V_{C1} = V_{C2} = V_{C3} = V_C$$

$$V_C = V_{in} - V_{out}$$

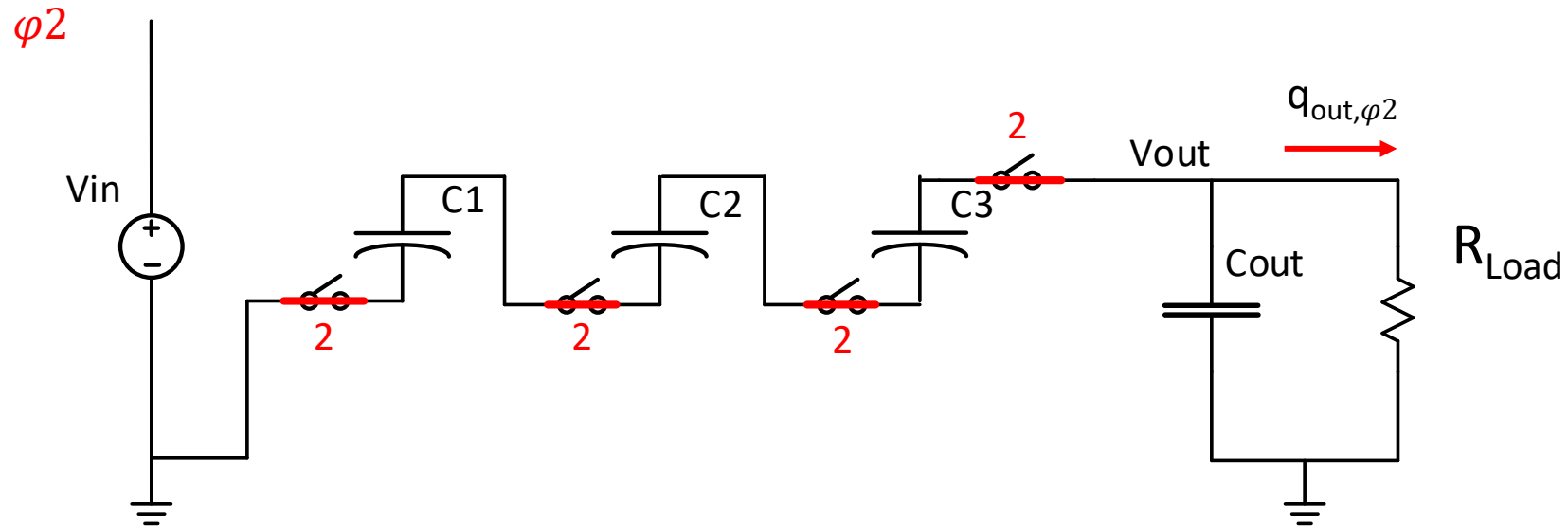
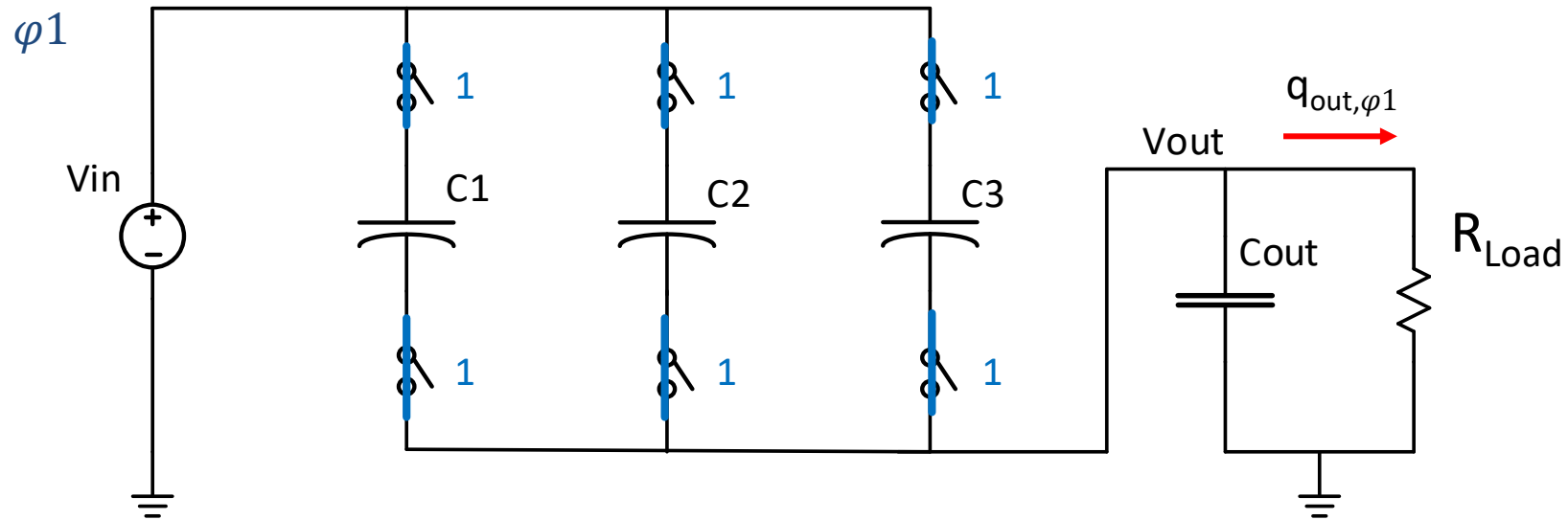


$$\phi 2: V_{out} = 3V_C = 3(V_{in} - V_{out})$$

$$V_{out} = \frac{3}{4} V_{in}$$

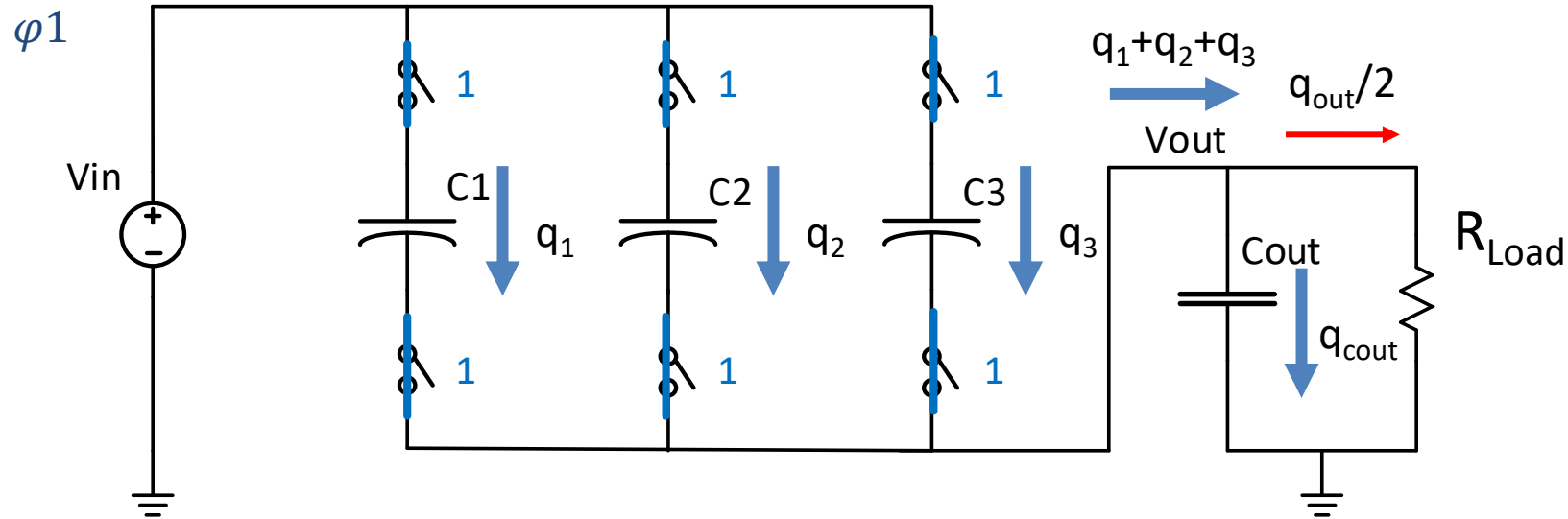
M=3/4 Series-Parallel SC Converter

Q5) How much charge flows through each capacitor in terms of q_{out} ?



$M=3/4$ Series-Parallel SC Converter

A5) How much charge flows through each capacitor in terms of q_{out} ?

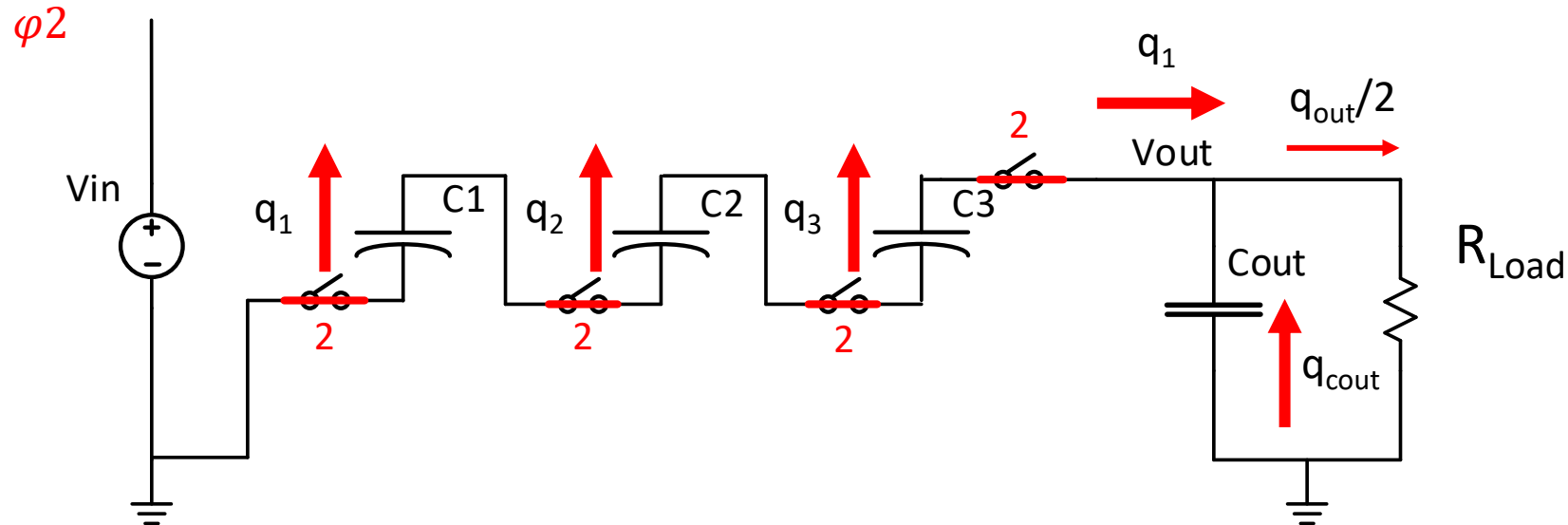


$$D = 50\% \Rightarrow q_{out, \phi 1} = q_{out, \phi 2} = q_{out}/2$$

$$q_{i, \phi 1} = q_{i, \phi 2} = q_i \text{ in steady state}$$

$$q_{in} = q_1 + q_2 + q_3$$

$$\phi 1: q_1 + q_2 + q_3 = \frac{q_{out}}{2} + q_{cout}$$



$$\phi 2: q_1 = q_2 = q_3$$

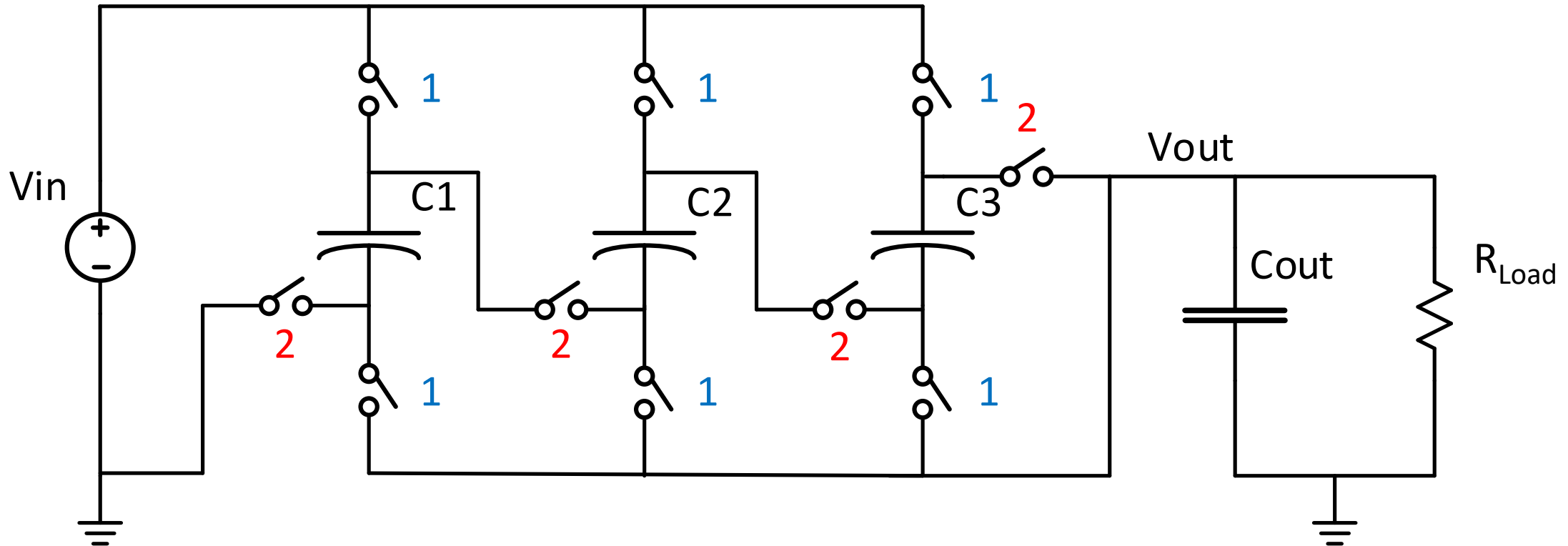
$$q_1 = \frac{q_{out}}{2} - q_{cout}$$

$$q_1 = q_2 = q_3 = \frac{q_{out}}{4}; q_{cout} = \frac{q_{out}}{4}$$

$$q_{in} = \frac{3q_{out}}{4}$$

M=3/4 Series-Parallel SC Converter

Q6) Simulate 3/4 Series-Parallel SC Converter for $V_{in}=4V$, $I_{out}=1mA$, $f_{sw}=4MHz$, $C1=C2=C3=1nF$, $R_{ON}=2\Omega$. What is the simulated R_{out} value?



A6)
$$R_{out} = \frac{3-2.953V}{1mA} = 47\Omega$$

